Setup

Where do we get Spark?

- The official website: ☆ Apache Spark™ - Unified Engine for large-scale data analytics
- Prerequisites (already done, students do not need to do this):
  - Install Java, specifically "openjdk11": `sudo apt install openjdk-11-jdk`
  - Set Spark driver binding address: `sudo hostname -b 127.0.0.1`
- We are interested in getting PySpark, currently, we have a way to easily install it via PyPi (Python Package Index, a.k.a pip): `pip install pyspark`

Nutanix VM

- The computer engineering department has a private cloud called "Nutanix".
- A virtual machine (VM), in this case, usually operates on a Linux system. Therefore, we can access it via secure shell (SSH).
- Credential:
  - <cmu_id>@10.10.184.13
  - password: iotbd2026

⚠ Change password after first login by using the command `passwd`

Crash course: SSH

Connect to a shell:

```
ssh pargorn_p@10.10.184.113
```

Using an SSH key (optional)

```
ssh-keygen
```

VS Code - remote connection

Connect to a remote server using VS Code

- Instruction: https://code.visualstudio.com/docs/remote/ssh
- Demo

Create a Python environment

- Use the "Python" extension from Microsoft (https://marketplace.visualstudio.com/items?itemName=ms-python.python) to:
  - Create a virtual environment
  - Install dependencies

Hello world: PySpark

- `pip install pyspark`
- Create a file `hello.py`:

```python
from pyspark.sql import SparkSession

# Initialize Spark session
spark = SparkSession.builder.appName("HelloWorld").getOrCreate()

# Create a sample dataframe
data = [("Alice", 34), ("Bob", 45), ("Cathy", 29)]
columns = ["Name", "Age"]
df = spark.createDataFrame(data, columns)

# Show the dataframe
df.show()

# Stop the Spark session
spark.stop()
```

- `python hello.py`

Utility commands

- Get which `python` would run: `which python`, side note: `which` command works with others too.
- See resource utilization: `htop`

Local development

Follow instructions in this link → Install Pyspark on Windows, Mac & Linux

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VS Code on macOS troubleshoot

- After the connection has been established, VS code might ask for remote server's password.
- Detailed error message may show: `stderr> Permission denied, please try again.`.
- This is caused by too many profiles being added to the same host name.
- Go to `~/.ssh/config` and remove duplicated profiles.
- Login again.